

# Visualize-flow test – author fresh diagrams

Recreates the manual test (the "show me the flow" + "describe core" thread) plus robustness variants. Each page is a deterministic scene under the pixel-gated harness; the payloads match the shape the new backend skill returns.

## WHAT THIS PROVES

- **Fix A — visualize authors a fresh diagram.** Replaced the prose-only skill with a structured LLM tool-call returning {narration, kind, nodes, edges, title, subtitle}. Kinds: pipeline / sequence / graph / tree. The frontend (HermesDiagram) swaps the cached payload for the authored one for the lifetime of the skillResult, then restores.
- **Fix B — mirror sentinel for the logic-view cache.** A cached logic-view whose source\_hash matches the file-view row is a 'we fell back' mirror. The route now treats it as cache-miss (re-extract) on read and skips persistence on cold-path write. Self-heals the poisoned cache that produced the file-vs-logic-identical bug.
- **How to read the proofs.** Each page below renders a real scene from the deterministic Playwright harness (?scene=<name>), which seeds the store with the exact skillResult shape the new backend skill returns. The pixel-diff gate (maxDiffPixelRatio 0.01) guards regressions; baselines are committed.

## KNOWN FOLLOW-UPS (NOT BLOCKING)

- Radial layout: pipelines with  $\geq 6$  nodes overlap or clip — a kind-aware layout (left  $\rightarrow$  right for pipeline, hierarchical for tree, sequence-rail for sequence) would let the diagram match the kind hint visually too.
- Subtitle "Local / web / HF sources  $\rightarrow$  loader  $\rightarrow$  chunker  $\rightarrow$  matcher  $\rightarrow$  GGUF" is rendering as a breadcrumb-style row, which works but isn't intentional — give the authored-diagram subtitle its own style so it reads as a caption.
- No e2e browser test against the live backend yet — the scenes harness proves the wiring with mocked skill payloads; a real-LLM run against a fixture would catch quality drift in the authored output (separate effort).

**pnpm verify: ALL GREEN** · typecheck 4/4 · lint 0 fails · unit (incl. new visualize-skill.test.ts) green · scenes 57/57 (3 new visualize-flow scenes  $\times$  3 viewports + 16 prior  $\times$  3).

# visualize-flow-pipeline

YOU SAID

"Show me the flow when I am using PersonalForge."

AI NARRATED

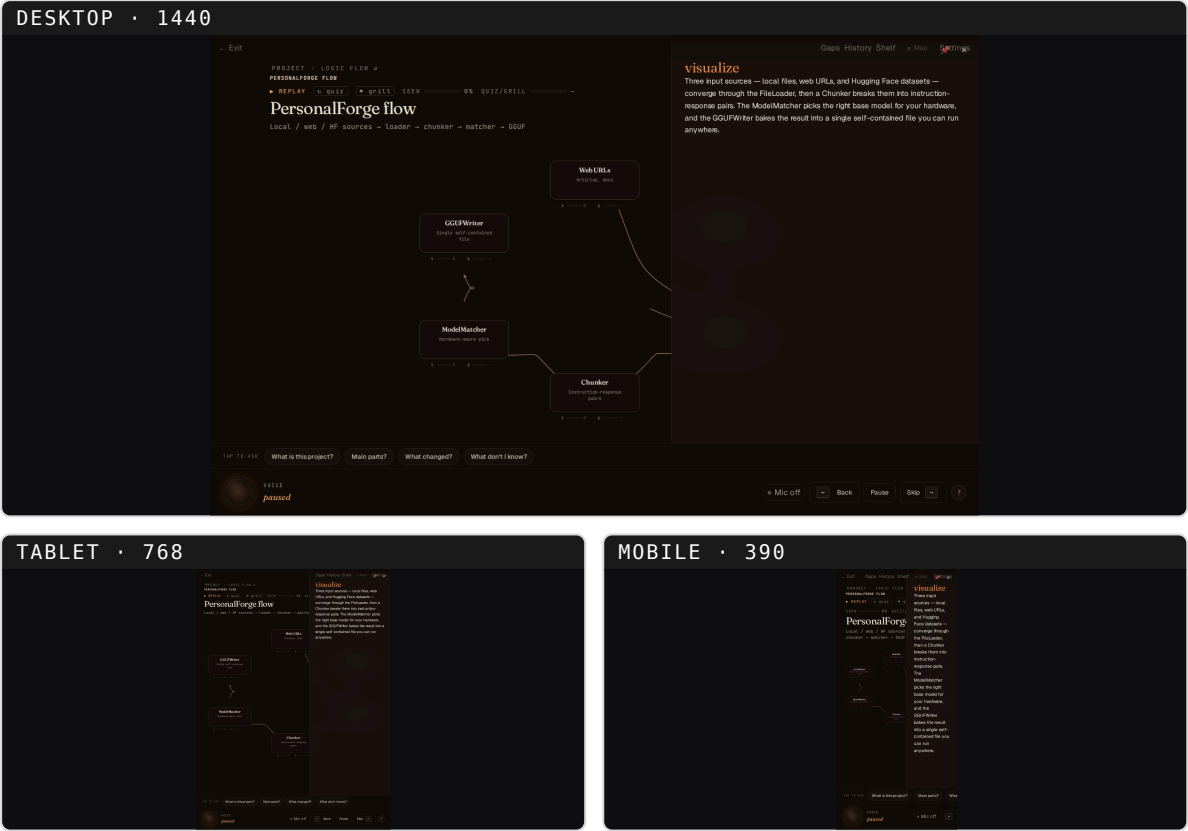
Three input sources — local files, web URLs, and Hugging Face datasets — converge through the FileLoader, then a Chunker breaks them into instruction-response pairs. The ModelMatcher picks the right base model for your hardware, and the GGUFWriter bakes the result into a single self-contained file you can run anywhere.

WHAT HAPPENED

visualize skill calls its `author\_diagram` LLM tool, returns a 7-node pipeline (local-files / web-urls / hf-datasets → file-loader → chunker → model-matcher → gguf-writer). HermesDiagram swaps the cached project map for this authored payload for the lifetime of the skill result, then restores. The spoken text refers to the exact node labels on screen — words and picture agree (the bug we set out to fix).

DIAGRAM KIND

pipeline



YOU SAID

"How do auth and the cache wire together?"

AI NARRATED

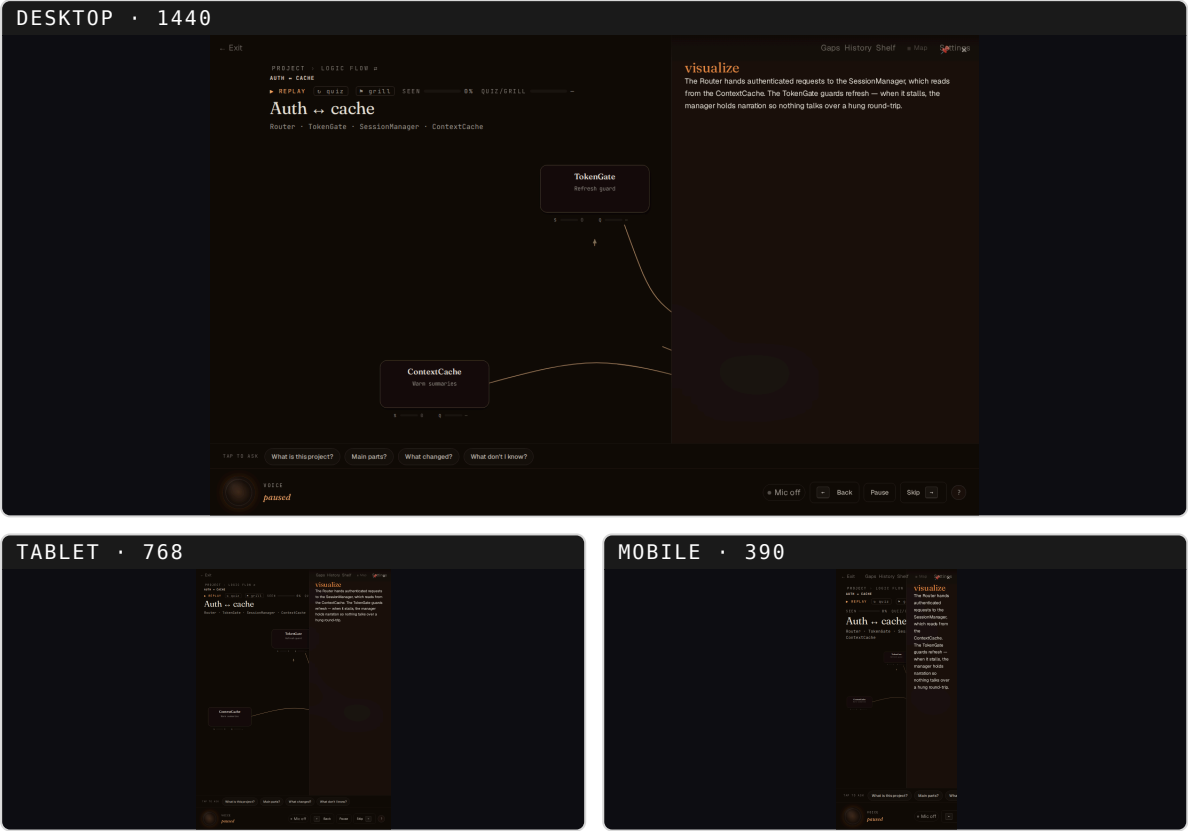
The Router hands authenticated requests to the SessionManager, which reads from the ContextCache. The TokenGate guards refresh — when it stalls, the manager holds narration so nothing talks over a hung round-trip.

WHAT HAPPENED

Robustness variant — same skill, different question shape. The LLM chose kind:`graph` (not pipeline), authored 4 nodes and 4 edges including a labelled "authed" edge. Proves the skill picks the right diagram kind for the question and the renderer handles non-pipeline shapes.

DIAGRAM KIND

graph



# visualize-flow-fallback

YOU SAID

(any visualize query, when the structured LLM call fails)

AI NARRATED

Couldn't author a custom diagram this time — but the project map you're on still tells the structural story: Core handles git + intelligence, Frontend renders the room and the diagram, Shared carries the types, and Voice gates barge-in.

WHAT HAPPENED

Voice north-star — when structured-call throws, visualize falls back to plain prose with NO authored diagram in the payload. The frontend swap is a no-op (authoredPayload === null) so the cached project map stays on screen. Spoken UX degrades gracefully; nothing goes silent or blank.

